Value Differencing Halves 4-Bit Quantization Damage in Attention

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Abstract

I began with a representational question: if neighboring token values become redundant with depth, should deep attention transmit what changed rather than repeat the raw value? I tested a one-line modification that subtracts the previous token's value projection with a coefficient that increases from zero to one across layers. It did not improve full-precision language modeling. At 20,000 matched training steps, the modified 27M-parameter models finished 0.0025–0.0066 BPB behind standard attention. The unexpected result appeared after weights-only 4-bit quantization: value differencing reduced W4 damage by 18.9% and 22.3% in two seeds, leaving the quantized models 0.0098 and 0.0178 BPB ahead. I then traced the effect to the architecture itself. Quantizing only attention weights cut damage by about half, and the value projection accounted for roughly 81% of that gap in both seeds. RMSE-matched Gaussian perturbations of attention weights produced the same ordering in all six paired controls, showing that the result is not specific to round-to-nearest quantization. The current implementation trains 6.7% slower and does not win at matched wall clock, so I claim a small-scale result about lower local sensitivity of attention parameters, not a production-efficiency result.

1 Introduction

Architecture studies usually stop at full precision. Deployment does not. Weights may later be rounded to four bits, pruned, or transformed by a calibrated quantizer, and two models that look equivalent before that step need not remain equivalent afterward.

This paper began somewhere else. I was studying adjacent-token redundancy in the values carried by attention. The measurements suggested a simple idea: deeper layers might transmit local changes more effectively than repeated raw content. I encoded that idea as *depth-scheduled value differencing*.[1] Early layers use ordinary values; each deeper layer subtracts more of the preceding token's value projection; the final layer transmits the full difference.

The full-precision result was unremarkable. Across the two final 20K runs, value differencing was slightly worse than standard attention. The interesting result emerged only after I rounded the large attention and MLP weight matrices to signed 4-bit values. That operation reversed the model ordering in both seeds: the modified checkpoints lost less quality and finished better after quantization.

That reversal could have been a measurement artifact or a property of some unrelated module. I therefore treated it as a question, not a conclusion. A component-by-component evaluation localized the gap to attention weights and, more specifically, to the value projection changed by the architecture. Quantization error itself was nearly unchanged, yet its effect on validation loss was roughly halved. Random attention-weight perturbations matched to the same relative RMSE produced the same result. Together, these controls support a precise conclusion: in this setting, training with value differencing makes the model less sensitive to perturbations of its attention parameters.

The scope is deliberately narrow. I study two training seeds, 27M-parameter FineWeb models, one tokenizer, one MLP family, BPB, and a calibration-free per-row W4 baseline. I do not claim a faster model, a production quantizer, or generalization to larger scales. The contribution is the localized empirical result and the sequence of controls that reduced a broad architectural idea to something the evidence actually supports.

[1] Earlier project artifacts use the name "DG Attention.' ' I use the descriptive term *value differencing* in this paper.

2 Value Differencing

2.1 Architecture

Let $x_{\ell,t}$ be the hidden state at layer ℓ and token position t. Standard attention projects it into queries, keys, and values. I leave the query and key paths unchanged and replace only the value payload:

$$p_{\ell,t} = W_V^{(\ell)} x_{\ell,t}, \qquad \widetilde{p}_{\ell,t} = p_{\ell,t} - \alpha_\ell p_{\ell,t-1}, \qquad \alpha_\ell = \frac{\ell}{L-1}.$$

Here $W_V^{(\ell)}$ is layer ℓ's learned value-projection matrix, $p_{\ell,t}$ is the value vector that standard attention would transmit, and α_ℓ is a fixed coefficient controlling how much of the previous token's projected value is subtracted. The tilde marks the modified payload.

The first token uses $p_{\ell,-1} = 0$. With $L = 11$, α_ℓ increases linearly from 0 to 1. Layer 0 is therefore standard attention; the deepest layer transmits the full previous-token difference. Attention then computes

$$y_\ell = W_O^{(\ell)}\ \mathrm{SDPA}\Big(Q_\ell, K_\ell, \widetilde{P}_\ell\Big).$$

Here Q_ℓ and K_ℓ are the usual query and key tensors, $W_O^{(\ell)}$ is the learned output projection, and SDPA is causal scaled dot-product attention.

This change preserves all tensor dimensions and adds no learned parameters to the matched baseline. Both models use the same grouped-query attention, RMSNorm, RoPE, Q-gain, and output projection. The implementation projects the values once, shifts the projected tensor by one token, and subtracts the scaled shift. It remains on the fused SDPA path and requires no custom attention kernel.

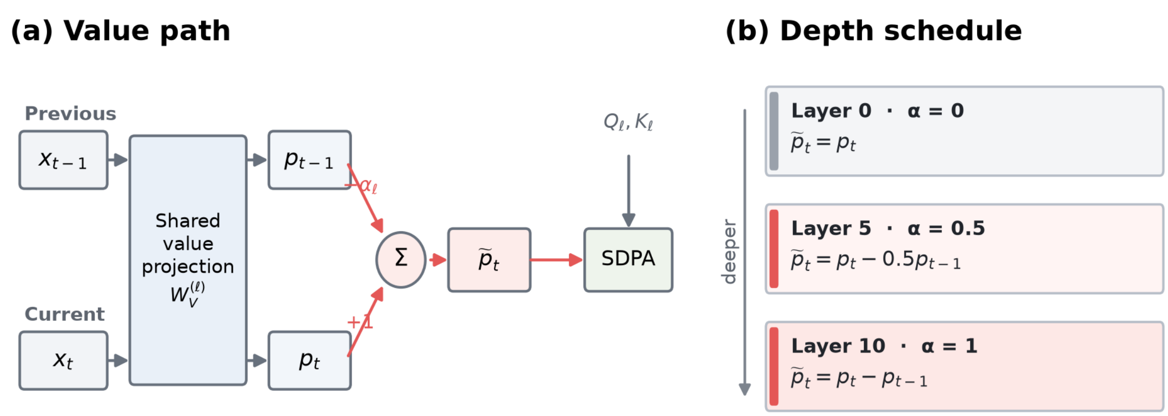

Figure 1: Value differencing changes only the payload supplied to attention. The same learned value projection processes the current and previous positions. The fixed coefficient α_ℓ grows with depth, moving from raw values to full previous-token differences; queries and keys are unchanged.

2.2 Quantization protocol

My primary diagnostic is weights-only, per-row, round-to-nearest W4. For each eligible weight row w, I compute

$$s = \frac{\|w\|_\infty}{7}, \qquad q = \mathrm{clip}(\mathrm{round}(w/s), -8, 7)\,, \qquad \widehat{w} = sq.$$

The canonical policy applies this operation symmetrically to large attention and MLP matrices in both architectures. Activations remain high precision; embeddings, normalization parameters, learned scales, and small control tensors are not quantized. I define quantization damage within each checkpoint:

$$\Delta_{\mathrm{W4}} = \mathrm{BPB}(\widehat{\theta}) - \mathrm{BPB}(\theta).$$

Lower BPB and lower damage are better. This is intentionally a transparent diagnostic, not a substitute for GPTQ, AWQ, or a production W4A16 kernel.

2.3 Training and evaluation

I train matched standard and value-difference models on FineWeb with a 1,024-token vocabulary. Each model has dimension 512, 11 skip-connected blocks, eight query heads, four key/value heads, ReLU-squared MLPs, and about 27M parameters. Runs use 49 data shards, sequence length 1,024, 393,216 tokens per optimizer step, Muon for matrix parameters, and 20,000 steps. The learning rate is constant through step 17K and then decays linearly for 3K steps.

All claim-bearing endpoint and localization results use raw pre-SWA checkpoints and the same certified 256-sequence evaluator. Comparisons are paired by training seed and checkpoint. The two independent endpoint seeds are 1337 and 42; intermediate checkpoints and perturbation seeds are not counted as additional training seeds. Appendix D records evaluator calibration, eligibility corrections, preregistration hashes, and artifact provenance.

3 Quantization Reverses the Ordering

3.1 Endpoints

Figure 2 contains the central result. Standard attention is better before quantization in both seeds. After W4, value differencing is better in both. It reduces W4 damage by 18.95% and 22.33%, overcoming the full-precision deficit and improving post-W4 BPB by 0.00978 and 0.01782. Appendix A reports the exact evaluator values.

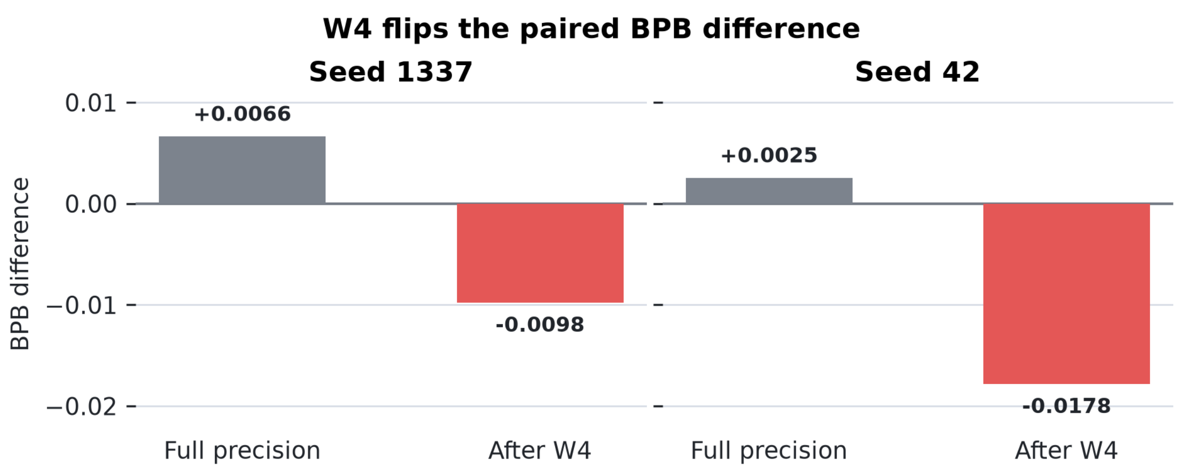

Figure 2: Paired BPB difference between value differencing and standard attention. Positive values mean standard attention has lower BPB; negative values mean value differencing has lower BPB. W4 changes the sign in both training seeds.

The comparison is matched by training tokens, not time. Peak VRAM is tied, and the measured implementation takes 6.72% longer per step. At the nearest saved approximately matched-wall-clock checkpoints, it wins only one of four late comparisons (Appendix A). The result is therefore not compute-adjusted superiority.

3.2 The gap persists through training

The seed-42 run saved 13 paired checkpoints from 2K to 20K steps. Value differencing takes less W4 damage at every checkpoint under the original conservative eligibility policy (Figure 3). These points are one trajectory, not 13 replications, but they show that the endpoint is not a single-checkpoint accident. The independent seed-1337 endpoint agrees in direction.

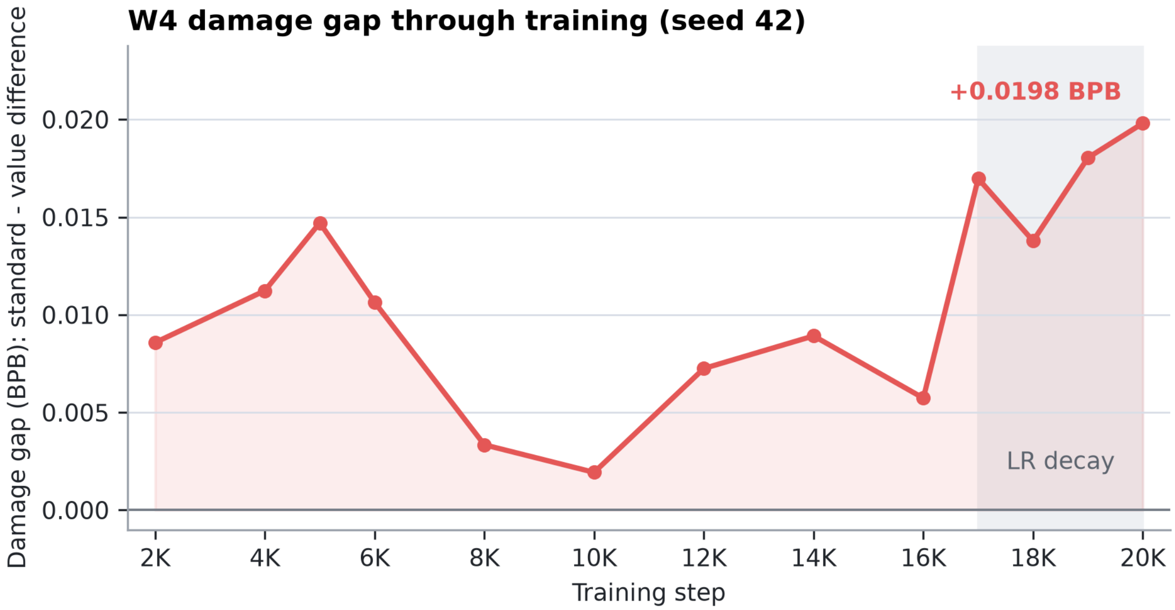

Figure 3: Standard minus value-difference W4 damage through the seed-42 run; positive values favor value differencing. The gap widens during the final learning-rate decay, from 0.00573 BPB at 16K to 0.01981 at 20K. I treat this as descriptive context, consistent with prior work on training dynamics and quantization [3, 7], not as a mechanism.

4 The Difference Lives in Attention

4.1 Component isolation

Quantizing one parameter group at a time gave a sharp answer: attention-only W4 reproduces or exceeds the entire architecture gap, whereas MLP-only W4 is null or weakly favors standard attention.

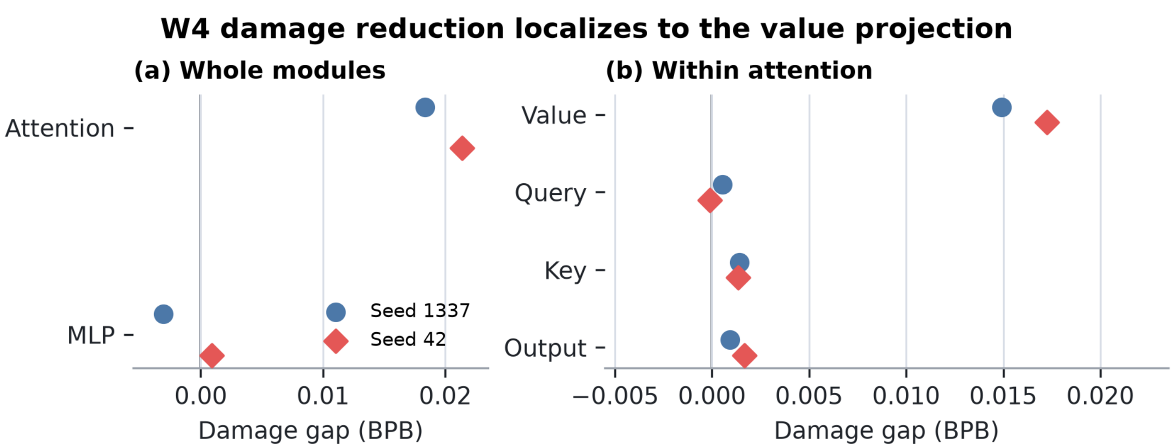

Figure 4: Standard-minus-value-difference damage by parameter group; positive values favor value differencing. The whole-module comparison is shown separately from the projections nested inside attention. Each point is a separate intervention, so the effects need not add linearly.

The value projection alone accounts for 81.4% and 80.7% of the matched attention-only gap. Isolating queries, keys, or the output projection explains less than 9% in either seed, and one query gap changes sign. The result thus localizes first to attention and then to the exact projection modified by value differencing. Exact paired damages and row provenance appear in Appendix B.

This is not simply lower quantization error. Attention-weight relative RMSE is 0.11854 versus 0.11759 in seed 1337 and 0.11717 versus 0.11691 in seed 42: a 0.2–0.8% difference in reconstruction error accompanies a 51.9–52.8% difference in BPB damage. The same-sized numerical error has a different functional consequence.

4.2 A perturbation control

Round-to-nearest W4 has a particular error structure. To test whether the ordering depends on that structure, I add independent Gaussian noise to eligible attention weights at per-tensor relative RMSE 0.117. Across three paired noise seeds per training seed, every comparison favors value differencing. Mean damage falls from 0.02310 to 0.00987 BPB in seed 1337 and from 0.02266 to 0.00960 BPB in seed 42 (Figure 5).

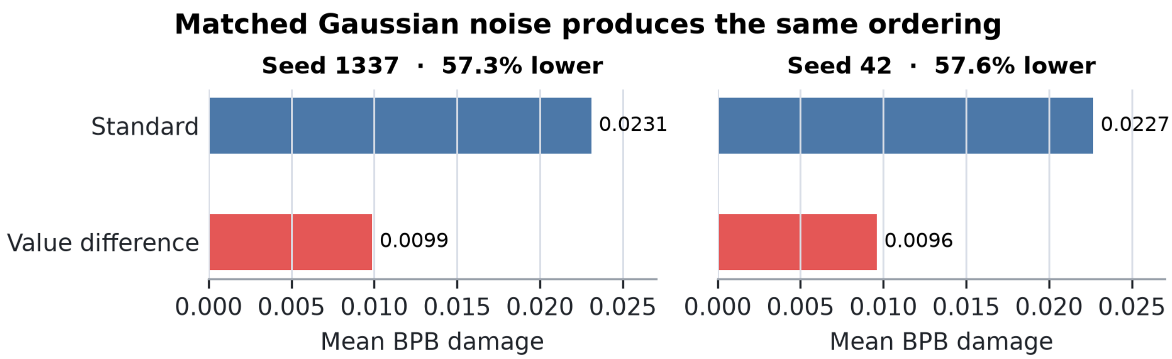

Figure 5: Mean BPB damage under RMSE-matched Gaussian perturbations of attention weights. Value differencing reduces damage by 57.3% and 57.6% across the two training seeds. Appendix C reports all six paired noise realizations.

This control rejects an explanation specific to deterministic W4 rounding. It supports lower local sensitivity within the tested attention-parameter subspace. It does **not** establish that the entire model lies in a globally flatter loss basin. Appendix C reports all six perturbation rows and the small historical eligibility asymmetry, which biases against the modified model.

5 What Did Not Survive

The final claim is narrower than the project that produced it. That narrowing is evidence, not backstory to hide.

I first expected value differencing to save memory or improve full-precision quality. It did neither: tensor dimensions and peak VRAM are unchanged, and the current implementation is slower. Early pruning and dead-zone results suggested broad compression robustness, but those rankings conflicted across seeds at 20K. I then preregistered a training-maturity inversion and a cross-module compensation hypothesis; the seed-42 trajectory and branch interventions rejected both. Gentle int6 quantization is effectively a null for both models.

What survives those tests is specific and repeatable within the measured setting: W4 damage is lower, the difference is attention-local, most of it lies in the value projection, and RMSE-matched random perturbations reproduce the ordering. Appendix E gives the complete hypothesis ledger, including the tests that failed.

6 Interpretation

The experiments separate *error size* from *error consequence*. W4 changes the two models' attention weights by nearly the same relative amount, but the change hurts standard attention roughly twice as much. Because the random-noise control agrees, the phenomenon is broader than one rounding rule. The strongest statement I can make is that value differencing lowers local sensitivity to attention-parameter perturbations in these checkpoints.

Why remains open. Value differencing may reduce the leverage of particular channels, change activation outliers, distribute payload information across neighboring positions, or guide optimization toward a less sensitive region of the attention subspace. One measured correlate points toward an outlier story: in a separate branch audit, a standard-model layer-3 MLP branch reached RMS 175.4 and maximum magnitude 134,060, versus 22.0 and 8,213 under value differencing. That measurement is not causal and did not come from the attention-W4 intervention, so I do not use it to explain the result.

7 Related Work

Architecture-side quantizability has direct precedent. Quantizable Transformers changes attention to suppress activation outliers and enable W8A8 quantization [2]. My result differs in both the intervention and regime: value semantics rather than softmax/no-op behavior, and weights-only W4 rather than full INT8. The shared lesson is that attention architecture can change what later compression destroys.

GPTQ and AWQ use calibration data to repair weight quantization [4, 5]; QuaRot and SpinQuant use rotations to reduce outliers [1, 6]. They are stronger deployment baselines than the transparent RTN diagnostic used here. Whether they erase or compound the architecture gap is unanswered.

KV Shifting Attention establishes learned local shifts of keys and values as prior art [9]. I isolate a fixed, subtractive V-side schedule and study parameter sensitivity rather than claiming token shifting itself as new. Differential Transformer is separate: it subtracts attention maps on the score side, whereas I difference value payloads across token positions [10]. Massive-activation work motivates the outlier hypothesis but does not establish it here [8].

Finally, recent studies show that quantization behavior changes over training [3, 7]. My trajectory agrees with that broad observation. The new result is the paired architecture difference and its localization to the modified value projection.

8 Limitations

This is a small-scale study: 27M parameters, FineWeb/SP1024, ReLU-squared MLPs, BPB evaluation, and two independent 20K training seeds. I report both seeds separately and do not claim statistical significance or a population-level effect. The quantizer is calibration-free, per-row RTN W4. I have not tested GPTQ, AWQ, production group formats, activation quantization, downstream tasks, or a real W4A16 kernel.

The implementation is 6.72% slower per step, ties peak VRAM, and gives no consistent advantage for equal elapsed training time. Nothing here supports an inference-speed claim or production adoption. A deployment study would require larger models, modern matched-byte quantizers, downstream evaluation, and kernel-level latency and memory measurements.

The experiments establish localization and local sensitivity, not mechanism. The result may disappear with scale, a different MLP family, or a quantizer that repairs the vulnerable directions in standard attention. Those are useful boundaries for future work, not assumptions I treat as passed.

9 Conclusion

Value differencing did not deliver the full-precision win I originally sought. It did something more specific: it made the attention parameters, especially the value projection, substantially less sensitive to W4 and matched random perturbations. In two 20K seeds, that difference was large enough to reverse the post-W4 model ordering.

The practical question remains open because the experiment is small and the implementation is slower. The scientific result is nevertheless clear within its scope: changing what attention transmits can change how much damage its weights cause when perturbed, even when their numerical reconstruction error is almost the same.

10 Data and Code Availability

I maintain the code at `https://github.com/ddavidgao/parameter-golf`. The submission release will pin the evaluator, run scripts, source archives, checkpoint hashes, raw logs, CSV summaries, cost ledgers, and SHA-256 manifests to an immutable commit. Every paid run used a software cost cap, durable local mirroring, and pod deletion after checksum verification.

A Runtime and Probe-Specific Results

The canonical endpoint values underlying Figure 2 are:

Seed	Model	FP BPB	W4 damage	Post-W4 BPB
1337	standard	1.1539035	0.0866231	1.2405266
1337	value difference	1.1605306	0.0702112	**1.2307418**
42	standard	1.1571615	0.0911300	1.2482915
42	value difference	1.1596927	0.0707769	**1.2304696**

These are raw pre-SWA checkpoints evaluated on 256 sequences under the symmetric W4 policy.

At nearest saved approximately matched-wall-clock checkpoints, value differencing wins one of four late comparisons. These pairs are not schedule-matched reruns and may occupy different learning-rate phases.

Standard step	Value-difference step	Time difference	Standard post-W4	Value-difference post-W4	Winner
17K	16K	-0.24%	1.31044	1.30170	value difference
18K	17K	+0.18%	1.28715	1.29583	standard
19K	18K	+0.60%	1.27167	1.27536	standard
20K	19K	+1.13%	1.24775	1.25639	standard

The broad compression hypothesis also failed. At 20K, MLP pruning and int5 dead-zone rankings disagree between the two seeds:

Seed	State / evaluator	Probe	Standard damage	Value-difference damage	Gap
1337	post-SWA, legacy	50% MLP prune	0.12995	0.16549	-0.03554
42	pre-SWA, 256 seq.	50% MLP prune	0.16704	0.15640	+0.01064
1337	post-SWA, legacy	MLP int5 dead zone	0.03215	0.07859	-0.04644
42	pre-SWA, 256 seq.	MLP int5 dead zone	0.04649	0.03939	+0.00709

Positive gaps favor value differencing. Because checkpoint states and evaluator scopes differ across the legacy seed-1337 and later seed-42 diagnostics, I interpret only paired within-row orderings and make no pruning claim.

B Evaluator and Component Details

The canonical endpoint policy quantizes both architectures' eligible final-layer K matrices. The original trajectory policy retained `blocks.10.attn.c_k` but not the modified model's equivalently placed `c_dk`; it therefore quantized one extra modified-model matrix. Two later symmetric controls, retaining both or quantizing both, preserve the ordering in both seeds.

The attention+MLP and attention-only rows below come from the symmetric quantize-both control log. The remaining rows come from the adjudicated component session. Every damage value is paired with the FP baseline from its own session. Cross-session FP reproduction agrees within 0.000011 BPB.

The full component split underlying Figure 4 is:

Seed	Component	Standard damage	Value-difference damage	Gap	Share of attention gap
1337	attention + MLP	0.086623	0.070211	+0.016412	–
1337	attention	0.035336	0.016992	+0.018344	100.0%
1337	MLP	0.044402	0.047444	-0.003042	–
1337	query	0.002503	0.001966	+0.000537	2.9%
1337	key	0.003459	0.002044	+0.001415	7.7%
1337	value	0.021857	0.006924	+0.014933	81.4%
1337	output	0.006117	0.005187	+0.000930	5.1%
42	attention + MLP	0.091130	0.070777	+0.020353	–
42	attention	0.040447	0.019077	+0.021370	100.0%
42	MLP	0.046621	0.045742	+0.000879	–

Seed	Component	Standard damage	Value-difference damage	Gap	Share of attention gap
42	query	0.002252	0.002357	-0.000104	-0.5%
42	key	0.003153	0.001804	+0.001349	6.3%
42	value	0.024911	0.007672	+0.017239	80.7%
42	output	0.006572	0.004887	+0.001685	7.9%

Shares use the canonical attention-only gap. They do not sum to 100% because each row quantizes one component in isolation and the interventions interact nonlinearly. An unpreregistered embedding-only diagnostic also favors value differencing in both seeds (+0.00546 and +0.00413 BPB), but embeddings are excluded from the canonical W4 policy and the observation carries no claim.

C Gaussian Attention-Weight Controls

Each row uses per-tensor relative RMSE 0.117. Noise seeds are paired across architectures.

Training seed	Noise seed	Standard damage	Value-difference damage	Gap
1337	1001	0.025228	0.010417	+0.014811
1337	1002	0.022408	0.009817	+0.012591
1337	1003	0.021659	0.009389	+0.012270
42	1001	0.023731	0.009222	+0.014510
42	1002	0.020759	0.009590	+0.011169
42	1003	0.023489	0.009980	+0.013509

The historical final-K eligibility asymmetry gives the modified model one additional perturbed matrix. The control is therefore conservative rather than exactly matrix-count matched.

D Evaluation and Preregistration Audit

The 256-sequence evaluator was calibrated against full evaluation on four seed-1337 endpoints and three perturbations. Two independent preflights found absolute damage differences of 0.000009–0.002298 and 0.000014–0.002255 BPB while preserving every architecture ranking. The later component implementation reproduced established W4 damage within 0.000043 BPB. Repeated checkpoints were bit-exact within the final control session; cross-session FP evaluation agreed within 0.000011 BPB.

Preregistration evolved only after recorded falsifications. The seed-42 source archive is `6116d694...f4dd`; its preregistration snapshot is `2d2c3bcb...6c2f`. The component archive is `61729e98...17d`; its later pre-run snapshot is `46f72fbe...6b5e`. I use *preregistered* only for tests and interpretation rules present before their corresponding runs. W4 became the headline through sequential narrowing; it was not the project's original hypothesis.

E Hypothesis Ledger

Hypothesis	Decisive test	Verdict
Value differencing reduces peak VRAM	Matched runtime measurements	Rejected: dimensions and VRAM tied
Value differencing is generally compression robust	int6, Gaussian, pruning, and dead-zone probes	Rejected: probe-specific
Training maturity reverses the architecture ranking	Preregistered seed-42 2K–20K trajectory	Rejected: inversion did not replicate
Attention differencing moves dependence into MLPs	Layerwise and branch-intervention gates	Rejected: predicted interaction absent
Value differencing reduces W4 damage	Two endpoints and one paired trajectory	Supported within scope
The W4 gap localizes to attention	Component isolation	Supported in both seeds
The attention gap localizes to the value projection	Q/K/value/output isolation	Supported in both seeds
Value differencing tolerates matched attention noise	Three noise seeds per checkpoint pair	Supported within tested subspace

References

[1] Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian Croci, Bo Li, Martin Jaggi, Dan Alistarh, Torsten Hoefler, and James Hensman. QuaRot: Outlier-free 4-bit inference in rotated LLMs. *arXiv preprint arXiv:2404.00456*, 2024.

[2] Yelysei Bondarenko, Markus Nagel, and Tijmen Blankevoort. Quantizable transformers: Removing outliers by helping attention heads do nothing. In *Advances in Neural Information Processing Systems*, 2023.

[3] Albert Catalan-Tatjer, Niccolò Ajroldi, and Jonas Geiping. Training dynamics impact post-training quantization robustness. *arXiv preprint arXiv:2510.06213*, 2025.

[4] Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023.

[5] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and acceleration. In *Proceedings of Machine Learning and Systems*, 2024.

[6] Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. SpinQuant: LLM quantization with learned rotations. *arXiv preprint arXiv:2405.16406*, 2024.

[7] Xu Ouyang, Tao Ge, Thomas Hartvigsen, Zhisong Zhang, Haitao Mi, and Dong Yu. Low-bit quantization favors undertrained LLMs: Scaling laws for quantized LLMs with 100t training tokens. *arXiv preprint arXiv:2411.17691*, 2024.

[8] Mingjie Sun, Xinlei Chen, J. Zico Kolter, and Zhuang Liu. Massive activations in large language models. *arXiv preprint arXiv:2402.17762*, 2024.

[9] Mingyu Xu, Wei Cheng, Bingning Wang, and Weipeng Chen. KV shifting attention enhances language modeling. *arXiv preprint arXiv:2411.19574*, 2024.

[10] Tianzhu Ye, Li Dong, Yuqing Xia, Yutao Sun, Yi Zhu, Gao Huang, and Furu Wei. Differential transformer. *arXiv preprint arXiv:2410.05258*, 2024.